

Alan Zheng

📍 Seattle, WA



// EXPERIENCE

SDE II @ Amazon Web Services

Aug 2023 – Present

IAM Data Plane • Seattle, WA

- ▶ Led migration of authentication traffic across 11 global regions to a sharded data architecture with zero customer impact.
- ▶ Unblocked a month-stalled production launch by standing up mutual-TLS service-to-service communication in one week, migrating the transport layer to a Netty-based stack with automated certificate rotation.
- ▶ Built a security scanner daemon that proactively detects data-consistency vulnerabilities, surfacing 50+ issues; now a mandatory release gate.
- ▶ Built core authentication APIs for a next-generation distributed auth system in Rust, reaching full feature parity with legacy Java services.
- ▶ Cut auth-API latency ~45% through systematic profiling and allocation, I/O, and framework-overhead optimization; stood up isolated load-test environments that gate releases on performance regressions.
- ▶ Saved 40+ operational hours annually with an AWS Bedrock-powered tool translating natural language into complex database queries.

SDE Intern @ Amazon Web Services

Summers 2021 & 2022

IAM Data Plane • Seattle, WA

- ▶ Built a CLI debugging tool for on-call engineers to diagnose IAM policy evaluation issues in production.
- ▶ Designed a DynamoDB-backed bootstrap rate limiter with metrics and deployment scripts to reduce maintenance load during fleet scaling.

Teaching Assistant & Tutor @ UVA Engineering

Sep 2021 – May 2023

Charlottesville, VA

- ▶ Graded exams and held office hours for the cryptocurrency technologies course; tutored linear algebra, vector calculus, machine learning, and data structures.

// PROJECTS

InterviewCoach

Python • LangGraph • AWS Bedrock • Multi-Agent

Agentic mock-interview app: a LangGraph orchestrator runs the interview while a parallel evaluator swarm scores each answer per dimension. Built on AWS Bedrock (Claude Sonnet + Haiku) with a swappable vLLM backend and a Streamlit UI.

Multiplayer Tetris with RL

Python • PyTorch • Deep Q-Learning • MARL

Trained multi-agent RL agents with Deep Q-Learning, shared parameters, and joint state modeling in a custom multiplayer Tetris environment. Achieved 87% win rate vs. a single-agent baseline.

Rust Ops CLI

Rust • AWS S3 • CLI • Ops Tooling

Rust CLI for pipeline health monitoring, performance benchmarking with S3 storage and run comparison, host scanning, and deployment diagnostics. Used daily by the team.

Standup Slack Bot

Python • Slack API • DynamoDB • Lambda

Multi-channel Slack bot for standup management with channel registration, admin controls, weekday-only rotation scheduling, and configurable announcements.

Drone Swarm Research

Python • RL • ROS • Vicon

Reinforcement learning model controlling a swarm of rescue drones in unknown environments. Produced a research paper and a live drone demonstration.

// SKILLS

Languages Rust, Java, Python, JavaScript, C++, Bash, SQL

AI / ML PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, spaCy, NumPy, Pandas

Cloud & Infra AWS (IAM, Lambda, EC2, S3, DynamoDB, Bedrock), Docker, GitHub Actions, Jenkins

Security & Crypto TLS / mTLS, SigV4, ECDSA, HMAC, X.509, cipher suites

Performance CodeGuru, perf, flamegraphs, heap & latency profiling

// EDUCATION

University of Washington • MS, Computer Science & Engineering • GPA 4.0

Expected Jun 2027

University of Virginia • BS, Computer Science • GPA 3.98 • Minors: Applied Math, Data Science

May 2023